

# VLM-Assisted Object-Goal Navigation with YOLO and Nav2 for Indoor Mobile Robots

A hybrid indoor mobile robot navigation project that combines **YOLO-based object detection**, the **ROS 2 Nav2 stack**, and a lightweight **Vision-Language Model (VLM)** to let a robot search for and approach objects described in natural language, such as *"the grey bottle on the table"*.

## Overview

Classical navigation systems can move a robot safely to coordinates, but they do not understand natural-language object descriptions. Closed-vocabulary detectors such as YOLO can recognize known classes quickly, but they cannot reliably match free-form descriptions or disambiguate between multiple similar objects on their own. This project adds a lightweight VLM as a semantic verification layer on top of a standard YOLO + Nav2 pipeline so the robot can interpret object descriptions more naturally while staying compatible with practical, resource-constrained hardware.

## Problem Statement

The project addresses indoor **object-goal navigation** where a user specifies a target object in natural language and the robot must search for it, verify that the detected object matches the description, and navigate to a safe standoff position without collision. The core thesis question is whether a lightweight external VLM, used sparingly as a semantic helper, can improve real-world object search compared with a YOLO-only baseline.

## Objectives

- Accept a natural-language object description as input.
- Detect candidate objects using YOLO or OAK-D spatial detections.
- Use a lightweight VLM to semantically verify whether an image crop matches the user's query.
- Use semantic cues such as presence and coarse direction (left, center, right) to guide search.
- Navigate safely using the ROS 2 Nav2 stack.
- Evaluate the hybrid system against a YOLO-only baseline on a real robot.

## System Architecture

The architecture is deliberately modular and resource-aware. YOLOv8 or an OAK-D SpatialDetection pipeline generates 2D or 3D object proposals, Nav2 performs global and local motion planning, and a lightweight VLM

running on an external machine verifies candidate regions of interest (ROIs) against the user query. The VLM is not used for low-level control; instead, it acts as a semantic assistant that helps decide whether a candidate object is the correct target and where the robot should bias its search next.

## Pipeline

1. User provides a natural-language object description.
2. Robot captures RGB observations from onboard sensors.
3. YOLO/OAK-D produces candidate detections.
4. Candidate bounding boxes are filtered and cropped into ROIs.
5. The external VLM checks whether the crop matches the query and may return coarse directional guidance.
6. A search policy chooses scan direction or next motion step.
7. Nav2 sends the robot toward a safe target pose.

## Key Contributions

According to the thesis, the main contributions of the project are:

- **Lightweight VLM semantic verification** for object-goal navigation using ROI-based semantic filtering rather than full scene reasoning.
- **Resource-aware hybrid architecture** that keeps classical perception and navigation in the loop while querying the VLM with a cooldown mechanism to manage compute and network cost.
- **Sector-based search policy** that combines detector outputs, semantic feedback, and robot pose to choose scan directions and short-range goals.
- **Real-robot evaluation** comparing a YOLO-only baseline with the proposed VLM-assisted configuration in indoor experiments.

## Experimental Setup

The system was evaluated on a differential-drive robot in indoor environments using three target categories: **chair**, **bottle**, and **person**. The thesis reports 10 trials per object category for each method, giving 30 trials for the YOLO-only baseline and 30 trials for the VLM-assisted configuration. The reported metrics include task outcome, detection time, reach time, number of turns, and number of moves.

## Results

The results show that semantic verification can help in some cases but introduces trade-offs.

| Target | YOLO-only success | VLM-assisted success |
|--------|-------------------|----------------------|
| Chair  | 60%               | 90%                  |
| Bottle | 70%               | 50%                  |
| Person | 90%               | 90%                  |

The chair category benefited the most from VLM assistance, suggesting that semantic verification is especially useful when multiple same-class objects must be disambiguated. Person performance remained strong in both systems, while bottle performance dropped with VLM assistance, likely because small targets and stricter semantic filtering made acceptance harder. The thesis also notes that detection time increased in the VLM-assisted system because of external semantic verification and network delay.

## Strengths

- Preserves the reliability of the classical Nav2 stack for safe motion.
- Adds semantic grounding without replacing the full navigation pipeline.
- Uses VLM calls selectively, making the approach more practical for constrained robots.
- Demonstrates real-robot results instead of simulation-only evaluation.

## Limitations

The thesis explicitly limits the work to single-object indoor navigation in relatively small office-like environments. It does not address long-horizon task planning, multi-step instructions, multi-robot coordination, or lifelong semantic mapping. It also reports that the external VLM introduces added latency and that small objects such as bottles remain challenging.

## Requirements

The report describes a system built around a mobile robot with onboard sensing, YOLO-based perception, a ROS 2 Nav2 navigation stack, and a lightweight VLM running externally over Wi-Fi. Before handing over the project, document the exact versions of ROS 2 packages, model weights, launch files, camera drivers, and any external inference dependencies used in your implementation.

A good handover should include at least:

- ROS 2 distribution and package list.

- YOLO model version and weights.
- OAK-D / camera setup instructions.
- Navigation parameters and maps.
- External VLM service setup.
- Network configuration between robot and external machine.
- Launch commands for baseline and VLM-assisted modes.

## How to Run

Replace the placeholders below with your actual package names and launch files before publishing.

```
# On the robot

cd YOUR_REPO

# Build workspace
colcon build
source install/setup.bash

# Launch robot stack
ros2 launch diff_drive_bridge no_cam_bringup.launch.py

# Launch perception/navigation pipeline
ros2 launch Urdf_model_description no_oak_mapless.launch.py

# Run VLM_commander

python3 vlm_commander.py

# Start external VLM service on host machine
# Open Terminal
# Download qwen.2.5vl:3B using OLLAMA

export OLLAMA_HOST=0.0.0.0

ollama serve
```

## Future Improvements

Based on the thesis scope and discussion, natural next steps include:

- Improving performance on small objects such as bottles.
- Reducing semantic verification latency.
- Extending from single-object commands to multi-step task instructions.
- Adding stronger semantic memory or mapping.
- Improving robustness under network variability.